COSMO® Cube — Hardware Computer Design Report

Session Report: COSMO-HARDWARE-REPORT-v1.md
Classification: Internal — Engineering + Strategic
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Date: 2026-06-12
Status: v1.0 — First complete hardware design session

Executive Summary

This report documents the complete design session for the COSMO® Cube — a hardware computer physically connected to the LOT Systems platform at lot-systems.com. The device is a flat, 40×40×5mm stainless steel ambient intelligence node: it receives pager-style notifications from the LOT AI engine, logs behavioral snapshots via a single Copy button, and captures environmental data through weather and AI-grade sensors.

The device is:
- A physical extension of the LOT operating system
- A behavioral signal collector (every button press is a data point)
- A hardware milestone for the COSMO® product line
- Designed for a 100-unit production run via PCBWay

7 documents produced this session. All sub-documents are in docs/hardware/.

Device Identity

Field	Value
Product name	COSMO® Cube
Form factor	40mm × 40mm × 5mm flat square
Body	316L stainless steel, CNC machined, 2 parts
Side A (Back)	Mirror-polished #8 finish — LOT® engraved
Side B (Front)	Satin SS — display, camera, button
Weight	~28g
Connectivity	WiFi 802.11n, BLE 5.0
Charging	Qi wireless (5W, included pad)
Platform	LOT Systems — lot-systems.com
Production run	100 units
Target unit cost	~$110 (cost) / $349 (retail)
Manufacturer	PCBWay (PCB + SMT + CNC)

Design Decisions Log

1. Form Factor (4×4cm, 5mm)

Decision: Flat square, credit-card-like footprint.

Rationale: Minimal desk presence. Sits flat or pockets like a money clip. The 5mm height is aggressive — requires custom thin LiPo (2.5mm), 0.8mm PCB, flush camera module. Achievable with careful stack-up (see COSMO-DEVICE-SPEC-v1.md §2, §5).
Challenge: Camera lens height is the critical constraint. HM01B0 + M7 macro lens stacks to ~3.5mm — within budget with 0.3mm SS plates.

2. Main MCU: ESP32-S3

Decision: ESP32-S3-MINI-1U (Espressif).
Rationale: Native 2.4GHz WiFi for LOT API polling. Built-in BLE 5.0. Hardware vector acceleration for on-device AI inference (notification filtering). Proven ecosystem. PCBWay stocks for turnkey assembly. 8MB flash sufficient for firmware + OTA slots.
Considered: nRF5340 (lower power, BLE-only — would need separate WiFi chip), Raspberry Pi RP2350 (no native WiFi).

3. Display: SSD1327 1.0" Grayscale OLED

Decision: 128×128 grayscale OLED.
Rationale: Perfect square format matches device footprint. 16-level grayscale enables elegant typography (LOT brand: no decorative colors, system-default aesthetic). Lower power than color LCD. 1.6mm module thickness fits 5mm budget.
Why not e-ink: Too slow for real-time notification rendering and button feedback animations.

4. Camera: Himax HM01B0

Decision: Ultra-low-power CMOS imager, 320×320.
Rationale: 1.1mW at 30fps — critical for battery life. CSP package (2.45×2.45mm) fits tight board. Used in Arduino Nano 33 BLE Sense — proven embedded ecosystem. Face detection on-device (ESP32-S3 vector instructions) for Soul Sync gate.
Fallback: OV2640 2MP for prototyping (thicker, more proven).

5. Wireless Charging Only (No USB)

Decision: Qi wireless charging, no USB port.
Rationale: Aesthetic purity (no port breaks SS surface). IP54 integrity (no port to seal). Behavioral signal: charging with intention, not plugging in.
Engineering note: BQ51013B + 30mm coil + ferrite fits in 5mm stack. Full charge in 2.5 hours.

6. Copy Button !’ LOT Log Tab

Decision: Single physical button = single behavioral gesture.
Rationale: The Copy button is the hardware equivalent of a LOT journal entry. Press it, and your environmental context (temp, humidity, light, orientation) is logged to lot-systems.com/log with tag [COSMO® Cube]. Intentional. Deliberate. Logged.
Signal value: Every button press timestamps a moment. Over time, press patterns become a behavioral signal (when do you reach for the device? What were conditions?).

7. Weather + AI-Grade Sensors

Decision: BME280 (weather) + ICM-42688-P (IMU) + APDS-9960 (gesture/light).
Rationale: BME280 is industry-standard for temp/

humidity/pressure — direct weather data. ICM-42688-P is TDK's highest-grade consumer IMU — detects motion, orientation, tap-to-wake, sleep posture. APDS-9960 enables gesture dismiss (wave to clear notification) and auto-sleep in dark/ambient conditions. "AI-grade": All three sensors have on-device signal processing, FIFO buffers, and hardware motion classification — no MCU cycles needed for basic event detection.

8. 2-Part 316L Stainless Steel Body

Decision: 316L (marine grade) SS, CNC machined, PCBWay.
Rationale: 316L is the gold standard for portable electronics (Apple Watch Series 2+ uses 316L). Superior corrosion resistance vs 304. CNC machining from PCBWay delivers ±0.05mm tolerance — sufficient for flush camera, button, display cutouts.
Why 2 parts: Back plate is flat (simple mirror polish). Front bezel has all cutouts (complex machining). Separation enables different finish per side.

Architecture Diagram

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%          lot-systems.com          %
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%   GET /api/hardware/notifications %  !• AI sends: "Coffee time!"
%   POST /api/hardware/log          %  !’ Button press + sensor data
%   GET /api/hardware/firmware      %  !• OTA update check
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%   Log Tab !• displays [COSMO® Cube]%
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Documents Produced This Session

Document	Description	Path
COSMO-HARDWARE-REPORT-v1.md	This session report	docs/hardware/
COSMO-DEVICE-SPEC-v1.md	Complete device specification	docs/hardware/
COSMO-BOM-v1.md	Bill of Materials — 100 unit run	docs/hardware/
COSMO-FIRMWARE-v1.md	Firmware architecture + code	docs/hardware/
COSMO-SOFTWARE-API-v1.md	LOT API connector + backend	docs/hardware/
COSMO-MANUFACTURING-v1.md	PCBWay manufacturing guide	docs/hardware/
COSMO-CHARGER-SPEC-v1.md	Wireless charger spec	docs/hardware/

Print-ready PDF manuals (generated 2026-07-20 — see §Session Continuation below):

PDF	Source	Path
COSMO-HARDWARE-REPORT-v1.pdf	This report	docs/hardware/pdf/
COSMO-DEVICE-SPEC-v1.pdf	Device specification	docs/hardware/pdf/
COSMO-BOM-v1.pdf	Bill of materials	docs/hardware/pdf/
COSMO-FIRMWARE-v1.pdf	Firmware architecture	docs/hardware/pdf/
COSMO-SOFTWARE-API-v1.pdf	LOT API integration	docs/hardware/pdf/
COSMO-MANUFACTURING-v1.pdf	PCBWay manufacturing guide	docs/hardware/pdf/
COSMO-CHARGER-SPEC-v1.pdf	Wireless charger spec	docs/hardware/pdf/

Generated by scripts/generate_cosmo_hardware_pdfs.cjs (pdfkit, same tool family as scripts/generate-badge-codex-pdf.cjs).

Components Buying List — Summary

	Component	MPN	Supplier	Unit Cost	Total (110)
1	ESP32-S3-MINI-1U	ESP32-S3-MINI-1U-N8	Mouser	$3.80	$418
2	SSD1327 OLED 1.0"	ER-OLED013-1W	BuyDisplay	$5.50	$605
3	Camera HM01B0	HM01B0-AAA	ArduCam	$4.20	$462
4	Weather BME280	BME280	Mouser 828-BME280	$2.80	$308
5	IMU ICM-42688-P	ICM-42688-P	Mouser	$3.50	$385
6	Light APDS-9960	APDS-9960	Mouser 630-APDS-9960	$2.10	$231
7	Qi Rx IC BQ51013B	BQ51013BRHLR	Mouser 595-BQ51013BRHLR	$2.80	$308

8	Qi Rx Coil 30mm	WE 760308101	Mouser / Alibaba	$1.50	$165
9	PMIC BQ25892	BQ25892RTWR	Mouser	$2.60	$286
10	LiPo 280mAh 35×35×2.5mm	Custom	Grepow	$6.50	$715
11	Copy Button	EVQ-Q2C03W	Mouser	$0.25	$28
12	RGB LED	APTR3216ZGCK	Mouser	$0.20	$22
13	LDO 1.8V AP2112K	AP2112K-1.8TRG1	DigiKey	$0.30	$33
14	Passives (caps, res)	Various	Mouser	$3.00	$330
15	PCB (4-layer, PCBWay)	Custom	PCBWay	$3.50	$385
16	SMT Assembly	Turnkey	PCBWay	$12.00	$1,320
17	SS Enclosure (2-part)	Custom CNC	PCBWay CNC	$40.00	$4,400
18	Qi Tx Pad (charger)	OEM custom	Alibaba	$9.00	$900
19	Packaging (box + foam)	Custom	Alibaba	$4.00	$400
	Grand Total (incl. 15% contingency)				~$12,340

Roadmap

Phase 0 — Design (Complete: This Session)

- [x] Device specification
- [x] Bill of materials
- [x] Firmware architecture
- [x] LOT API integration design
- [x] Manufacturing guide
- [x] Charger specification
- [x] Session report

Phase 1 — Engineering (Weeks 1–4)

- [ ] PCB schematic capture (KiCad 8.0)
- [ ] PCB layout (35×35mm, 4-layer)
- [ ] Enclosure CAD (Fusion 360 or FreeCAD)
- [ ] Gerbers + DXF/STEP generated
- [ ] LOT backend: hardware API endpoints coded

Phase 2 — Prototype (Weeks 5–7)

- [ ] 10-unit prototype order (PCBWay)
- [ ] Firmware v0.1: boot + WiFi + display
- [ ] Firmware v0.2: API polling + notification display
- [ ] Firmware v0.3: Copy button + sensor logging
- [ ] Hardware validation: all sensors, charging, camera

Phase 3 — Production (Weeks 8–10)

- [ ] 100-unit production order (PCBWay)
- [ ] Factory firmware flash + provisioning
- [ ] QA: all 100 units (checklist in COSMO-MANUFACTURING-v1.md)
- [ ] Packaging + shipping

Phase 4 — Launch

- [ ] LOT web app: My Devices page
- [ ] LOT Log tab: hardware entry display
- [ ] Notification push: QI-46 Engine !' device
- [ ] OTA infrastructure for firmware updates
- [ ] FCC/CE certification (pre-commercial, budget ~$20K)
- [ ] Retail listing at $349 (Purple+ Benchmark tier required)

PDF Manual Plan — Status

v1.0 planned six audience-specific manuals via Pandoc or Figma. As of the 2026-07-20 session, the seven source documents are instead rendered directly to PDF via pdfkit (scripts/generate_cosmo_hardware_pdfs.cjs) — no Pandoc dependency, no external tool, consistent with how this repo already produces the Badge Codex PDFs. One PDF per source document, not one per audience:

PDF (docs/hardware/pdf/)	Source	Pages produced
COSMO-HARDWARE-REPORT-v1.pdf	This report	24
COSMO-DEVICE-SPEC-v1.pdf	Device specification	24
COSMO-BOM-v1.pdf	Bill of materials	33
COSMO-FIRMWARE-v1.pdf	Firmware architecture	27
COSMO-SOFTWARE-API-v1.pdf	LOT API integration	18
COSMO-MANUFACTURING-v1.pdf	Manufacturing guide	18
COSMO-CHARGER-SPEC-v1.pdf	Charger spec	9

Not yet done: the audience-curated cuts (a 4-page Quick Start card, an A5 print-trimmed retail insert) are a design pass, not a content problem — they need the actual box/insert layout from COSMO-BOM-v1.md §10.2, which depends on the box vendor being selected. Tracked as a Phase 3 (Production) task, not blocking engineering.

Strategic Notes

Why this device first:
The COSMO® Cube is the simplest hardware node in the COSMO® product line. It is not a robot. It is not a companion AI. It is an ambient sensor + notification terminal + behavioral logger. It validates the hardware pipeline (PCBWay, firmware, LOT API integration) at low risk before COSMO® robotics.

Why 100 units:
100 units is the minimum for meaningful market feedback and the minimum for PCBWay CNC pricing to be reasonable on SS enclosures. It is below MOQ risk for all components. Under $15K total.

Fundable from current revenue.

The Copy button is the signal:
Every time a LOT user presses the Copy button, they are making a behavioral gesture — "I am here, this is my context, log it." Over months, these presses form a pattern: when do they reach for the device? What light levels? What temperature? What time of day? This is Soul Sync data. It is the hardware's contribution to the Quantum Intent Engine.

"Coffee time!" is the interface:
The LOT AI sends "Coffee time!" to the device. The user sees it on the mirror-finished square on their desk. They pause. They get coffee. The AI learned this pattern from their behavior. The hardware closes the loop — knowledge becomes action, action becomes habit, habit becomes identity.

Session Continuation — 2026-07-20

The v1.0 design session (below) ran on claude/brave-lamport-t9z5u8 and was
never merged — docs/hardware/ did not exist on master. This continuation session (claude/brave-lamport-xwycvi) carried the design forward:

- Ported all 7 documents into this branch's docs/hardware/ (they did not previously exist anywhere reachable from master).
- Corrected two inconsistencies found on read-through:
  - COSMO-CHARGER-SPEC-v1.md §5 described charging with the mirror-polished (coil) side facing *up* — that would put the Qi coil away from the pad. Fixed to mirror-side-down, display-up (see that document's §6 Revision Notes).
  - COSMO-SOFTWARE-API-v1.md §2.2 modeled the backend as SERIAL-keyed raw SQL, disconnected from how this app actually persists data (Sequelize + umzug migrations, UUID keys — see migrations/20240525154723_add-logs.cjs and src/server/models/log.ts). Rewrote the schema and route handlers to match, and dropped the redundant hardware_logs table — a Copy button press is just a row in the existing logs table (see that document's §7 Revision Notes).
- Generated the PDF manuals that v1.0 only planned (PDF Manual Plan section above) — 7 PDFs, 153 pages total, via a new pdfkit script (scripts/generate_cosmo_hardware_pdfs.cjs).
- Attempted to read brand.lot-systems.com, lot-systems.com/about, and institute.lot-systems.com/cqgs.html for brand-voice and CQGS context — all three returned HTTP 403 to this session's fetch tool. Brand voice and trademark conventions were instead drawn from in-repo sources (README.md, docs/corporate/LOT_ROBOTICS_COSMO.md, docs/benchmark/LOT-DOCTRINE.md), which already carry the same LOT®/COSMO® language.

No physical units exist yet — this remains a design-and-BOM deliverable, not a shipped device. Nothing was implemented in src/ this session: the API route and DB migration snippets in COSMO-SOFTWARE-API-v1.md are reference code for whoever builds Phase 1, not code running in the live app today.

Session Compression Summary

Session 1: Hardware Computer Design — COSMO® Cube
Date: 2026-06-12
Output: 7 documents, ~25,000 words, full hardware specification to production-ready
Key decisions: ESP32-S3 MCU, HM01B0 camera, SSD1327 OLED, 316L SS enclosure, PCBWay manufacturing, Qi wireless charging, BME280 + ICM-42688-P + APDS-9960 sensors
Branch: claude/brave-lamport-t9z5u8 (never merged)

Session 2: Continuation, correction, PDF manuals
Date: 2026-07-20
Output: 7 docs ported + corrected, 7 PDF manuals generated (153 pages), 1 generator script
Next action: PCB schematic in KiCad 8.0, LOT API hardware endpoints, merge docs/hardware/ to master
Branch: claude/brave-lamport-xwycvi

COSMO® CIA — LOT Systems, Inc.
Inventor: Vadim Marmeladov
Named for Kuzya Cosmo Marmeladov
Made in the USA.

"A flat square of polished steel that says 'Coffee time!' is not a gadget.
It is proof that the machine learned something true about you."
— Vadim Marmeladov